Formulas & Tricks — Made Easy

Units 3 & 4

VCE Mathematical Methods Units 3 & 4 · Functions · Trig · Transformations · Coordinate geometry · Calculus · Probability · Statistics · Algebra · Algorithms · Exam technique

This is the short, friendly version of the big formula sheet. Every rule comes with a picture, a table or a worked example. Look for the coloured boxes:

TRICK — saves time | TRAP — where marks are lost | CHECK — catch your own mistakes | EXAMPLE — see it worked out

1 Functions and graphs

Interval notation — square bracket or round bracket?

Write	Means	On a graph	Example
`[a, b]`	both ends included	● filled dots at both ends	`[1, 4]` → $1 \le x \le 4$
`(a, b)`	both ends left out	○ open circles at both ends	`(1, 4)` → $1 < x < 4$
`[a, b)`	only a included	● at a, ○ at b	`[1, 4)` → $1 \le x < 4$
`R \ {2}`	every real number except 2	hole / asymptote at x = 2	`(−∞, 2) ∪ (2, ∞)`

∞ always gets a round bracket — you can never "reach" infinity.

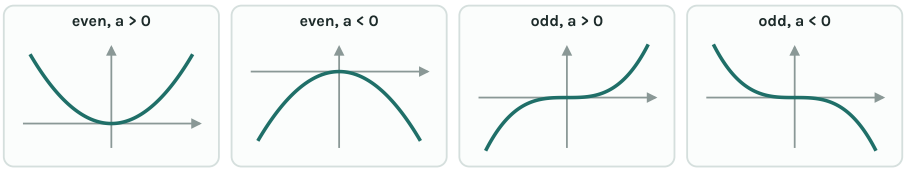
Maximal (largest possible) domain

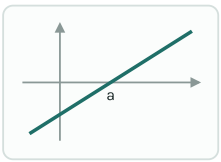
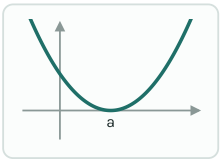
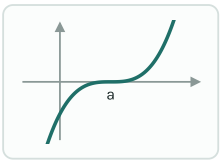
If the rule has…	You need…	Example	Domain
a square root √(g(x))	`g(x) ≥ 0`	`√(x − 3)`	`[3, ∞)`
a fraction 1 / g(x)	`g(x) ≠ 0`	`1 / (x + 5)`	`R \ {−5}`
a log $\log_e(g(x))$	`g(x) > 0`	$\log_e(2 - x)$	`(−∞, 2)`
more than one of these	all of them at once	see the example below	

EXAMPLE `y = √(x + 2) + 1/(x − 1)`. Root: x ≥ −2. Fraction: x ≠ 1. Together: **`[−2, 1) ∪ (1, ∞)`**.

Polynomials — shape from the degree

A polynomial of degree **n** has **at most n** x-intercepts and **at most n − 1** turning points. Only the **leading term** decides where the arms go:



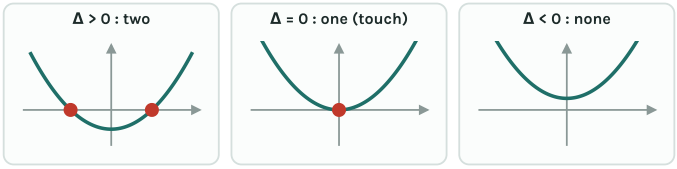
Factor	Picture near x = a	What the graph does	Sign changes?
$(x - a)$		**cuts** straight through	yes
$(x - a)^2$		**touches** and turns back (turning point)	no
$(x - a)^3$		**flattens** then goes through (stationary point of inflection)	yes

Rule of thumb: **even** power → touch, **odd** power → cross.

EXAMPLE $y = (x - 1)^2(x + 3)$: touches the x-axis at x = 1, cuts at x = −3. It is a cubic but has only **2 distinct** zeros.

Quadratics $y = ax^2 + bx + c$

What you want	Formula	Quick note
x of the turning point	$x = -b / (2a)$	then substitute to get y
turning point form	$y = a(x - h)^2 + k \rightarrow TP\ (h, k)$	watch the minus: $y = 2(x + 3)^2 - 1$ has TP (−3, −1)
quadratic formula	$x = (-b \pm \sqrt{(b^2 - 4ac)}) / (2a)$	leave surds exact
axis of symmetry	$x = (p + q) / 2$	halfway between the x-intercepts p and q
discriminant	$\Delta = b^2 - 4ac$	tells you **how many** solutions — see pictures



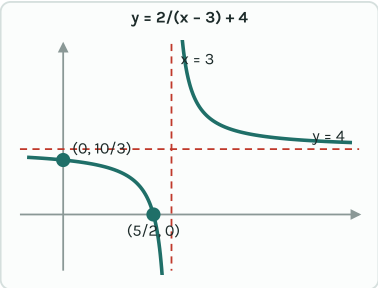
TRICK

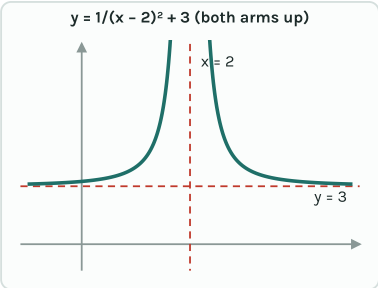
Use the discriminant — do not solve

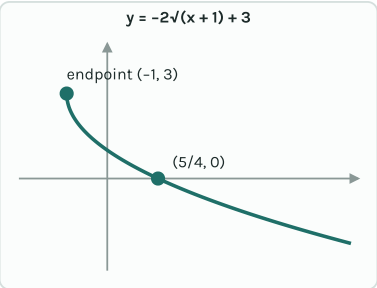
"For what k is the line a tangent?" → set the rules equal, collect into $ax^2 + bx + c = 0$, then put **$\Delta = 0$**. ($\Delta > 0$: two intersections, $\Delta < 0$: none.) No need to actually solve the quadratic.

Hyperbola, truncus and square-root graphs

Graph	Rule	Asymptotes / endpoint	Domain	Range
Hyperbola	$y = a/(x - h) + k$	x = h, y = k	$R \setminus \{h\}$	$R \setminus \{k\}$
Truncus	$y = a/(x - h)^2 + k$	x = h, y = k	$R \setminus \{h\}$	(k, ∞) if $a > 0$
Square root	$y = a\sqrt{(x - h)} + k$	endpoint (h, k)	$[h, \infty)$	$[k, \infty)$ if $a > 0$ $(-\infty, k]$ if $a < 0$







EXAMPLE `y = 2/(x − 3) + 4` x-intercept: 2/(x − 3) = −4 → x − 3 = −½ → x = 5/2. y-intercept: 2/(−3) + 4 = 10/3.

TRICK

Split the numerator instead of long division

Make the top contain the bottom: `(3x + 1)/(x − 2) = (3(x − 2) + 7)/(x − 2) = 3 + 7/(x − 2)`. Now you can read the asymptotes straight off: **x = 2** and **y = 3**.

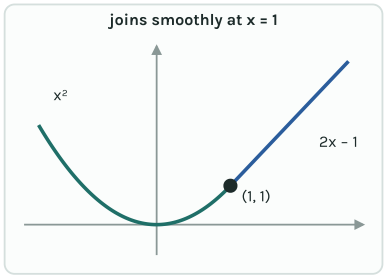
CHECK

Endpoint or asymptote? Pick the right bracket

A square-root endpoint **is** reached → square bracket: range of y = −2√(x + 1) + 3 is `(−∞, 3]`. An asymptote is **never** reached → round bracket / "\": range of the truncus above is `(3, ∞)`.

Hybrid (piecewise) functions

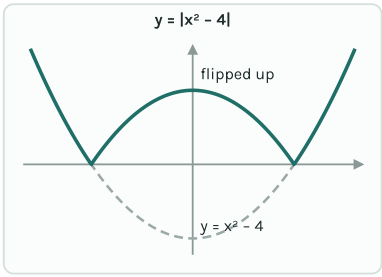
To be…	Check at the join x = a
continuous (no gap)	left value = right value = f(a)
differentiable (smooth, no corner)	continuous **and** left gradient = right gradient



EXAMPLE f(x) = x² for x ≤ 1, 2x − 1 for x > 1.
Values: 1² = 1 and 2(1) − 1 = 1 ✓ Gradients: 2x = 2 and 2 ✓ → smooth join.
If it were x² and x: values match (1 = 1) but gradients 2 ≠ 1 → a **corner**, not differentiable.

Modulus |x| and odd / even functions

Type	Rule	Example
equation	`|x − a| = b → x = a ± b`	`|x − 2| = 5 → x = 7 or −3`
"less than"	`|x| < a ⇔ −a < x < a`	`|x| < 3 → (−3, 3)`
"greater than"	`|x| > a ⇔ x < −a or x > a`	`two separate pieces`
graph of |f(x)|	flip everything **below** the x-axis up	
graph of f(|x|)	keep x ≥ 0, mirror it in the y-axis	



Even function	Odd function
`f(−x) = f(x)` — mirror image in the y-axis. e.g. x², cos x, |x|	`f(−x) = −f(x)` — same after a half-turn about the origin. e.g. x³, sin x, 1/x

Combining functions: sums, products and power functions

Combination	Rule
(f + g)(x), (fg)(x)	`domain = dom f ∩ dom g`
even × even, odd × odd	even
even × odd	odd
even + odd (e.g. $x^3 + x^2$)	usually **neither**

Power function	Domain	Range
`y = √x = x^(1/2)`	`[0, ∞)`	`[0, ∞)`
`y = ∛x = x^(1/3)`	`R`	`R`
`y = 1/x = x⁻¹`	`R \ {0}`	`R \ {0}`
`y = 1/x² = x⁻²`	`R \ {0}`	`(0, ∞)`

EXAMPLE dom(√x + √(2 − x)): √x needs x ≥ 0, √(2 − x) needs x ≤ 2. Overlap (∩, not ∪) → **`[0, 2]`**.

Tip: rewrite roots and fractions in index form (√x = x^½, 1/x² = x⁻²) before you differentiate or integrate.

TRAP

Double factor touches, triple factor flattens

At $(x - a)^2$ the curve turns back. At $(x - a)^3$ it flattens and **keeps going**. Students often draw a turning point at a triple factor — don't.

CHECK

Count features against the degree

A degree-n polynomial can't have more than n x-intercepts or more than n − 1 turning points. A quartic drawn with 4 turning points is wrong on sight.

Try it yourself — Section 1

1. Find the maximal domain of `y = √(4 − x) + 1/(x + 1)`.
2. Find the asymptotes of `y = (2x + 5)/(x + 1)`. (Hint: split the numerator.)
3. Describe what `y = (x + 2)²(x − 1)³` does at x = −2 and at x = 1.

2 Trigonometric functions

Radians and exact values

`π rad = 180°` · degrees → radians: × π/180 · radians → degrees: × 180/π

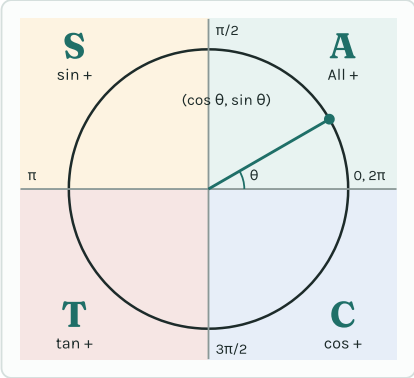
θ	0	π/6 (30°)	π/4 (45°)	π/3 (60°)	π/2 (90°)
sin θ	0	1/2	√2/2	√3/2	1
cos θ	1	√3/2	√2/2	1/2	0
tan θ	0	√3/3 (= 1/√3)	1	√3	undefined

TRICK

Remember the sin row in 5 seconds

Write `√0/2, √1/2, √2/2, √3/2, √4/2` → 0, ½, √2/2, √3/2, 1. The cos row is the **same numbers backwards**. tan = sin ÷ cos.

The unit circle and the quadrants (ASTC)



"All Students Take Calculus" — which ratio is positive

Quadrant	Angle in terms of base angle θ	Positive
1 (0 to π/2)	`θ`	all
2 (π/2 to π)	`π − θ`	sin only
3 (π to 3π/2)	`π + θ`	tan only
4 (3π/2 to 2π)	`2π − θ`	cos only

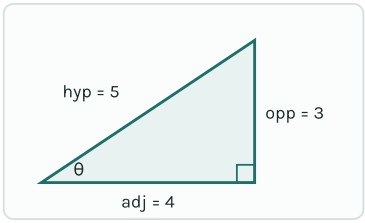
EXAMPLE sin(7π/6): 7π/6 = π + π/6 → quadrant 3, base angle π/6. sin is negative there, so **sin(7π/6) = −1/2**.

Also useful: `sin(−θ) = −sin θ`, `cos(−θ) = cos θ`, `sin(π/2 − θ) = cos θ`, `sin²θ + cos²θ = 1`, `tan θ = sin θ / cos θ`.

Finding one ratio from another

EXAMPLE sin θ = 3/5 and θ ∈ (π/2, π). Find cos θ and tan θ.

① Draw the triangle: opp 3, hyp 5 → adj 4 (3-4-5).
② Quadrant 2: cos is negative → **cos θ = −4/5**.
③ **tan θ = (3/5) ÷ (−4/5) = −3/4**.



the triangle gives the size, the quadrant gives the sign

TRAP

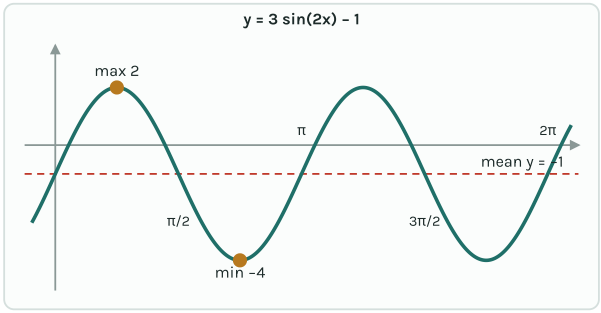
The identity gives the size, the quadrant gives the sign

cos²θ = 1 − 9/25 = 16/25 gives cos θ = ±4/5. Only the given domain tells you which sign. Writing +4/5 loses the mark even though the arithmetic is right.

Graphs of sine and cosine: $y = a\sin(n(x + b)) + c$

Feature	Formula	For $y = 3\sin(2x) - 1$
amplitude	$\lvert a\rvert$	3
period	$2\pi / \lvert n\rvert$	$2\pi/2 = \pi$
mean (centre) line	$y = c$	$y = -1$
range	$[c - \lvert a\rvert, c + \lvert a\rvert]$	$[-4, 2]$
cycles in $[0, 2\pi]$	$2\pi \div$ period	2 cycles

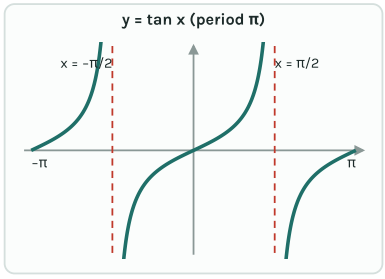
Key points are a **quarter-period apart**: sine starts at the mean line going up; cosine starts at a max (if $a > 0$).



The tangent graph $y = a\tan(n(x + b)) + c$

Feature	Rule
period	$\pi / \lvert n\rvert$
asymptotes of tan x	$x = \pi/2 + k\pi,\ k \in Z$
x-intercepts of tan x	$x = k\pi,\ k \in Z$
amplitude	**none** — range is R

EXAMPLE $y = \tan(2x)$: period $\pi/2$. Asymptotes: $2x = \pi/2 + k\pi \rightarrow x = \pi/4 + k\pi/2$.



Solving a trig equation on a restricted domain — 4 steps

Example: solve $2\sin(2x) + 1 = 0$ for $x \in [0, 2\pi]$.

1. Isolate the trig ratio
   $2\sin(2x) + 1 = 0 \rightarrow \sin(2x) = -\frac{1}{2}$
2. Change the domain
   $x \in [0, 2\pi] \rightarrow 2x \in [0, 4\pi]$
3. Base angle → quadrants → add periods
   base $\pi/6$; sin − in Q3, Q4: $7\pi/6$, $11\pi/6$, then $+2\pi$: $19\pi/6$, $23\pi/6$
4. Undo (÷ 2)
   $x = 7\pi/12, 11\pi/12, 19\pi/12, 23\pi/12$

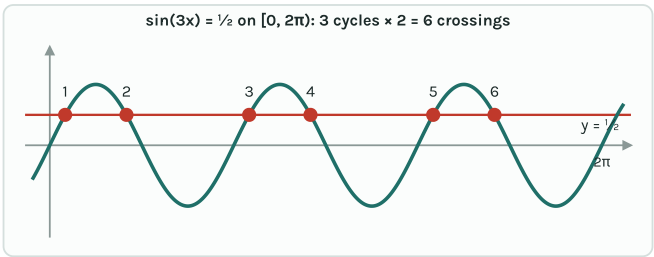
TRICK

Change the domain, not the answers

Convert the domain first ($2x \in [0, 4\pi]$), find **every** answer for $2x$ in that bigger interval, then divide by 2 at the very end. Solving for x first and adding periods later is where solutions go missing.

How many solutions should I expect?

Equation (over one interval of length 2π)	Number of solutions	Example on $[0, 2\pi)$
$\sin(nx) = k$ or $\cos(nx) = k$, with $\lvert k\rvert < 1$	**2n**	$\sin(3x) = \frac{1}{2} \rightarrow$ 6 solutions
$\sin(nx) = \pm 1$ or $\cos(nx) = \pm 1$	**n** (one per cycle)	$\sin(2x) = 1 \rightarrow$ 2 solutions
$\tan(nx) = k$ (any k)	**n per π**, so 2n per 2π	$\tan x = -\sqrt{3} \rightarrow$ 2 solutions
sin or cos = k with $\lvert k\rvert > 1$	**0**	$\sin x = 2 \rightarrow$ no solution



EXAMPLE sin(3x) = ½ on [0, 2π): 3x ∈ [0, 6π). Base angle π/6, sin + in Q1, Q2:
3x = π/6, 5π/6, 13π/6, 17π/6, 25π/6, 29π/6 → **x = π/18, 5π/18, 13π/18, 17π/18, 25π/18, 29π/18** — six answers, matching the graph. **If you have fewer, you dropped some.**

CHECK

Range check kills impossible equations

y = 3 sin(2x) − 1 only takes values in [−4, 2]. So `3 sin(2x) − 1 = 3` has **no solution** — check this before you start solving.

General solutions (n ∈ Z)

Equation	General solution	Example
`sin x = a`	`x = 2nπ + sin⁻¹(a) or x = (2n + 1)π − sin⁻¹(a)`	`sin x = ½ → x = 2nπ + π/6 or (2n+1)π − π/6`
`cos x = a`	`x = 2nπ ± cos⁻¹(a)`	`cos x = ½ → x = 2nπ ± π/3`
`tan x = a`	`x = nπ + tan⁻¹(a)`	`tan x = 1 → x = nπ + π/4`

CHECK

Put n = 0, 1, −1 back in

Always write **n ∈ Z**. Then substitute n = 0, 1, −1 — the values that land in the domain should match your restricted-domain answers.

TRAP

Factorise before you read the translation

`sin(2x − π/3)` is **not** a shift of π/3. Write it as `sin(2(x − π/6))`: the shift is **π/6 to the right**. Any time n ≠ 1, take n out of the bracket first.

TRICK

Double-angle formulas (handy, but optional in Methods)

`sin(2θ) = 2 sin θ cos θ · cos(2θ) = cos²θ − sin²θ = 1 − 2sin²θ = 2cos²θ − 1`. Methods questions are written so you can avoid these, so never rely on them.

Transforming trig graphs — what changes what

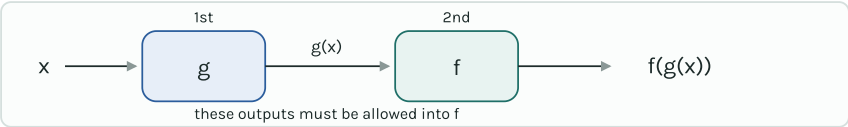
Change	Affects	Example
a (stretch up/down)	amplitude and range only	`y = −2 sin x: amplitude 2, flipped`
n (stretch sideways)	**period only**	`y = sin 3x: period 2π/3`
b (shift sideways)	neither period nor range	`y = sin(x − π/4): π/4 right`
c (shift up/down)	range and mean line	`y = sin x + 2: range [1, 3]`

Try it yourself — Section 2

1. How many solutions does `cos(2x) = 0.3` have for x ∈ [0, 2π]?
2. Solve `cos x = −½` for x ∈ [0, 2π].
3. State the amplitude, period and range of `y = −2 cos(3x) + 5`.

3 Composite functions and transformations

Composite functions f(g(x)) — think of two machines



Fact	Rule	What it means
Order	`f(g(x))`	do **g first**, then f (also written f ∘ g)
Exists when	`ran g ⊆ dom f`	everything g puts out must be allowed into f
Domain	`dom f(g(x)) = dom g`	after any restriction you had to make
Not symmetric	`f(g(x)) ≠ g(f(x))`	one can exist while the other does not

EXAMPLE — IT FAILS, THEN FIX IT $f(x) = \sqrt{x}$, $g(x) = x - 3$ (dom R).
ran g = R, dom f = [0, ∞). R ⊄ [0, ∞) → f(g(x)) **does not exist**.
Fix: restrict dom g to [3, ∞). Then ran g = [0, ∞) ✓ and $f(g(x)) = \sqrt{(x - 3)}$.

EXAMPLE — RANGE OF A COMPOSITE $f(x) = x^2$, $g(x) = x - 1$, dom g = [0, 4].
Step 1: ran g = [−1, 3].
Step 2: square that set → **[0, 9]**, **not** [1, 9], because x = 1 gives 0, the minimum.

TRAP

It is ran g ⊆ dom f, not dom g ⊆ dom f

The test is on the **outputs** of the inside function. Write down ran g, write down dom f, then compare those two sets.

The transformation rule y = a f(b(x + c)) + d

Letter	Effect	Exam wording	Example from y = f(x)
a	multiplies every **y** by a (a < 0 also flips)	dilation of factor |a| from the x-axis	`y = 3f(x): 3 × taller`
b	**divides** every **x** by b (b < 0 also flips)	dilation of factor 1/|b| from the y-axis	`y = f(2x): squashed, factor ½`
c	moves left by c	translation of c units in the negative x direction	`y = f(x + 3): 3 left`
d	moves up by d	translation of d units in the positive y direction	`y = f(x) + 2: 2 up`

Image of a point: `(p, q) → (p/b − c, aq + d)`. **Order:** dilations and reflections first, translations last.

Example: y = 2f(3(x − 1)) + 5 sends the point (2, 4) to …

Coordinate	Working	Result
x	divide by 3, then add 1 → 2/3 + 1	**5/3**
y	multiply by 2, then add 5 → 2 × 4 + 5	**13**
Check with $f(x) = x^2$: $2(3(5/3 - 1))^2 + 5 = 2(2)^2 + 5$		= 13 ✓

TRAP

Dilation factors from the y-axis go upside down

f(2x) is a dilation of factor ½ from the y-axis, not 2 — the graph is squashed towards the y-axis. Vertical ones read straight off: 2f(x) is factor 2 from the x-axis.

TRICK

Asymptotes and endpoints move too

Apply the point rule to x = h and y = k as well. For y = 1/x → y = 2/(x − 3) + 4, the asymptotes x = 0, y = 0 move to **x = 3, y = 4**. Fastest way to sketch any transformed graph.

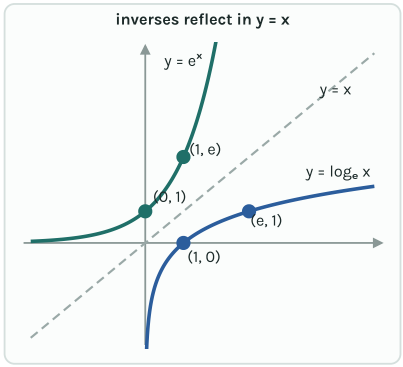
Describing transformations in words

Rule change	Correct exam wording
`y = 3f(x)`	dilation of factor 3 **from the x-axis** (= parallel to the y-axis)
`y = f(2x)`	dilation of factor ½ **from the y-axis** (= parallel to the x-axis)
`y = −f(x)`	reflection **in** the x-axis (not "about" or "across")
`y = f(−x)`	reflection in the y-axis
`y = f(x − 2)`	translation of 2 units in the positive direction of the x-axis

List dilations and reflections first, then translations. Matrix form (same information):
`[x′, y′] = [[1/b, 0], [0, a]]·[x, y] + [−c, d]`.

Inverse functions f^{-1}

Fact	Rule
Exists only if	f is **one-to-one** (passes the horizontal line test)
Find the rule	swap x and y, then solve for y
Domain and range	`dom f⁻¹ = ran f, ran f⁻¹ = dom f`
Graph	reflect y = f(x) in the line y = x
Points, asymptotes	(a, b) → (b, a); x = h ↔ y = h
Check	`f(f⁻¹(x)) = x`



f(x)	Swap and solve	$f^{-1}(x)$	dom f^{-1} (= ran f)
`2x − 5`	`x = 2y − 5`	`(x + 5)/2`	R
`(x − 1)³ + 2`	`x − 2 = (y − 1)³`	`(x − 2)^(1/3) + 1`	R
`e^(2x) − 1`	`x + 1 = e^(2y)`	`½ log_e(x + 1)`	`(−1, ∞)`
`x², x ≥ 0`	`x = y², y ≥ 0`	`√x`	`[0, ∞)`

TRAP

f^{-1} is NOT 1 ÷ f

The −1 is not a power. f(x) = x + 2 has $f^{-1}(x) = x - 2$, not 1/(x + 2). Same with $\sin^{-1}$ (inverse sine) vs 1/sin.

TRAP

Not one-to-one? Restrict the domain first

$f(x) = x^2 - 4x + 1$ turns at x = 2, so restrict to [2, ∞) or (−∞, 2]. Then recompute the range on that piece — it becomes the domain of f^{-1}.

TRICK

Where do f and f^{-1} meet?

If f is **increasing**, solve f(x) = x: e.g. √(x + 2) = x gives (2, 2). If f is decreasing they can also meet off the line y = x — then solve $f(x) = f^{-1}(x)$.

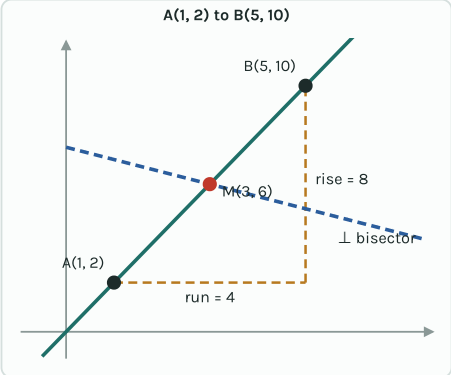
Try it yourself — Section 3

1. f(x) = √x and g(x) = x + 2 with dom g = R. Does f(g(x)) exist? If not, give the largest domain of g that fixes it.
2. Find the image of the point (1, 3) under $y = -f(2(x + 1)) + 4$.
3. Describe the transformations from y = sin x to $y = \sin(2x - \pi/3)$.
4. Find f^{-1} for f(x) = e^(x) + 3 and state its domain.

4 Coordinate geometry (straight lines)

Not a section on the official sheet — but used everywhere: tangents, normals, intersections, transformations.

Find	Formula	A(1, 2), B(5, 10)
Gradient	$m = (y_2 - y_1)/(x_2 - x_1)$	8/4 = **2**
Midpoint	$((x_1+x_2)/2, (y_1+y_2)/2)$	**(3, 6)**
Distance	$\sqrt{((x_2-x_1)^2 + (y_2-y_1)^2)}$	√80 = **4√5**
Line	$y - y_1 = m(x - x_1)$	y - 2 = 2(x - 1) → **y = 2x**



Lines are…	Gradients	Example with y = 2x + 1
parallel	$m_1 = m_2$	$y = 2x - 7$
perpendicular	$m_1 \times m_2 = -1$ (flip and change sign)	$y = -\frac{1}{2}x + 4$
horizontal / vertical	$m = 0 \to y = k$; undefined → $x = k$	$y = 3$, $x = -2$
angle with the x-axis	$\tan\theta = m$	$m = 1 \to \theta = 45°$

EXAMPLE Perpendicular bisector of AB: passes through M(3, 6) with gradient -½ (flip 2, change sign).
y - 6 = -½(x - 3) → **y = -½x + 7.5**.

EXAMPLE — INTERSECTION Where do y = 2x and x + y = 6 meet? Substitute: x + 2x = 6 → x = 2, y = 4 → **(2, 4)**.

TRICK

Intercepts in two seconds

For ax + by = c: cover the y-term → x-intercept c/a; cover the x-term → y-intercept c/b. 3x + 2y = 12 → (4, 0) and (0, 6).

Try it yourself — Section 4

1. Find the gradient, midpoint and length of the segment from P(-2, 1) to Q(4, 9).
2. Find the equation of the line through (2, -1) perpendicular to y = 3x + 5.

5 Exponentials and logarithms

Index laws (same base only)	
$a^m \times a^n = a^{m+n}$	$2^3 \times 2^4 = 2^7$
$a^m \div a^n = a^{m-n}$	$e^5 \div e^2 = e^3$
$(a^m)^n = a^{mn}$	$(x^2)^3 = x^6$
$a^0 = 1$, $a^{-m} = 1/a^m$	$2^{-3} = 1/8$
$a^{(m/n)} = (\sqrt[n]{a})^m$	$8^{(2/3)} = 2^2 = 4$

Log laws ($\log_e$ = ln)	
$\log_a y = x \Leftrightarrow a^x = y$	a log is an index
$\log(mn) = \log m + \log n$	$\ln 6 = \ln 2 + \ln 3$
$\log(m/n) = \log m - \log n$	$\ln 5 - \ln 2 = \ln 2.5$
$\log(m^p) = p \log m$	$\ln 8 = 3 \ln 2$
$\log 1 = 0$, $\log_a a = 1$	$e^{(\ln k)} = k$, $\ln(e^x) = x$
change of base	$\log_a b = \ln b / \ln a$

TRAP

There is no law for the log of a sum

`ln(m + n) ≠ ln m + ln n` and `ln m × ln n ≠ ln(mn)`. Laws only work on a product, quotient or power **inside one log**.

Solving exponential and log equations

Type	Method	Example
same base	equate the powers	`8^x = 2^(x+4) → 3x = x + 4 → x = 2`
one exponential	**isolate first**, then take ln	`2e^(3x) = 7 → e^(3x) = 7/2 → x = ⅓ ln(7/2)`
quadratic in e^x	let $A = e^x$, reject $A \le 0$	`e^(2x) − 5e^x + 6 = 0 → A = 2, 3 → x = ln 2, ln 3`
logs	collapse to one log, then "unlog"	`ln x + ln(x − 3) = ln 4 → x² − 3x − 4 = 0`

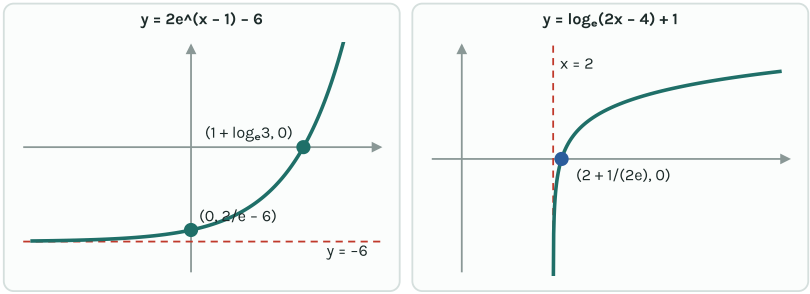
TRAP

Check every log answer in the ORIGINAL equation

$x^2 - 3x - 4 = 0$ gives $x = 4$ or $x = -1$. But $\ln(-1)$ is undefined, so **$x = 4$ only**. Likewise $2 \ln x = \ln 9$ gives $x = 3$ only, while $\ln(x^2) = \ln 9$ gives $x = \pm 3$.

Graphs of exponentials and logs

	$y = A\,e^{b(x+c)} + d$	$y = A \log_e(b(x+c)) + d$
Asymptote	horizontal **$y = d$**	vertical, where $b(x + c) = 0$
Domain	`R`	solve $b(x + c) > 0$
Range	`(d, ∞) if A > 0; (−∞, d) if A < 0`	`R`
Key point of the basic graph	`e^x: (0, 1), (1, e)`	`ln x: (1, 0), (e, 1)`



e^x and $\log_e x$ are inverses — see the reflection graph in Section 3. If A and d have the same sign, the exponential graph has **no x-intercept**.

Growth and decay

Model	Formula	Example: P = 500e^(0.04t)
growth / decay	`P = P₀e^(kt) (k > 0 grows, k < 0 decays)`	starts at 500
doubling time / half-life	`t = ln 2 / |k|`	ln 2 / 0.04 ≈ **17.3**
time to multiply by m	`t = ln m / k`	triples at ln 3 / 0.04 ≈ 27.5
find k from data	substitute a known point, then ln	P(10) = 800 → e^(10k) = 1.6 → k = ln 1.6 / 10

TRICK

Doubling time ignores the starting amount

P_0 cancels in $P_0e^{kt} = 2P_0$, so the doubling time is always $\ln 2 \div k$. Also: any base can be turned into e: `a^x = e^(x ln a)`.

Try it yourself — Section 5

1. Solve 3e^(2x) − 1 = 11, giving an exact answer.
2. Solve e^(2x) − e^x − 6 = 0.
3. State the asymptote and range of y = −3e^x + 2.

6 Differentiation

Standard derivatives

f(x)	f′(x)	Example
x^n	nx^{n-1}	$x^{-3} \to -3x^{-4}$
e^(kx)	k e^(kx)	e^(5x) → 5e^(5x)
$\log_e(kx)$	1/x	ln(3x) → 1/x
$\log_e(ax + b)$	a/(ax + b)	ln(2x+1) → 2/(2x+1)

f(x)	f′(x)	Example
sin(kx)	k cos(kx)	sin 2x → 2cos 2x
cos(kx)	−k sin(kx)	cos 3x → −3sin 3x
tan(kx)	k / $\cos^2(kx)$	tan x → $1/\cos^2 x$
a^x	$a^x \ln a$	$2^x \to 2^x \ln 2$

Calculus only works in **radians**. Rewrite √x as x^½ and $1/x^3$ as x^{-3} first.

The three rules

Rule	Formula	Example
Chain (function inside a function)	dy/dx = dy/du × du/dx	$(2x + 1)^5 \to 5(2x + 1)^4 \times 2 = 10(2x + 1)^4$
Product (u × v)	(uv)′ = u′v + uv′	x^2e^(3x) → 2xe^(3x) + $3x^2$e^(3x) = xe^(3x)(2 + 3x)
Quotient (u ÷ v)	(u/v)′ = (u′v − uv′)/v^2	x/(x + 1) → ((x + 1) − x)/$(x + 1)^2$ = $1/(x + 1)^2$

TRICK

Simplify before you differentiate

$(x^3 + 2x)/x = x^2 + 2$ → derivative 2x. No quotient rule needed. Log laws work too: $\ln(x^2\sqrt{x}) = (5/2) \ln x \to 5/(2x)$.

TRICK

Spot f′/f

If the top is the derivative of the bottom, it came from a log: d/dx $\ln(x^2 + 3x) = (2x + 3)/(x^2 + 3x)$.

TRAP

ln(kx) differentiates to 1/x — the k disappears

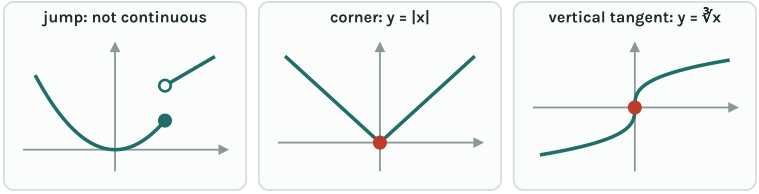
ln(kx) = ln k + ln x, and ln k is a constant. But ln(ax + b) → a/(ax + b), where the a stays. Also: quotient rule order is u′v − uv′ (swap it and every sign flips).

First principles and the meaning of f′

Idea	Formula
Derivative (first principles)	f′(x) = lim (h→0) [f(x + h) − f(x)] / h
Instantaneous rate = gradient of tangent	f′(a)
Average rate = gradient of chord	[f(b) − f(a)] / (b − a)
Second derivative	f″(x) = d^2y/dx^2 (NOT $(dy/dx)^2$)

EXAMPLE $f(x) = x^2 + 3x$ by first principles: $f(x + h) - f(x) = 2xh + h^2 + 3h$. Divide by h: 2x + h + 3. Let h → 0: **f′(x) = 2x + 3**.

Where is f NOT differentiable? Look for these three shapes



Differentiable ⇒ continuous, but continuous does **not** mean differentiable ($y = |x|$ is continuous at 0 but has a corner). For a hybrid function: check values match first, then gradients.

CHECK

Gradient of an inverse: use the image point

$(f^{-1})'(b) = 1 / f'(a)$ where $f(a) = b$. $f(x) = x^3 + 1$: $f(1) = 2$, $f'(1) = 3$ → $(f^{-1})'(\mathbf{2}) = 1/3$ (at $x = 2$, not $x = 1$).

Try it yourself — Section 6

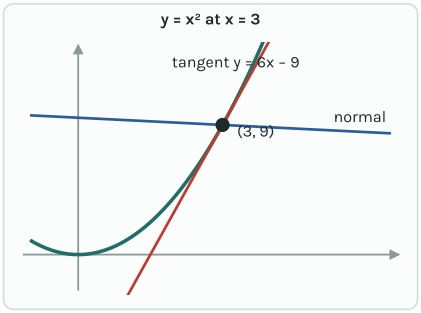
1. Differentiate: (a) $(3x - 2)^4$ (b) $x \sin x$ (c) $e^{(x^2)}$ (d) $\ln(5x)$
2. Differentiate $x^2 / (x + 1)$ using the quotient rule.

7 Applications of differentiation

Tangents and normals

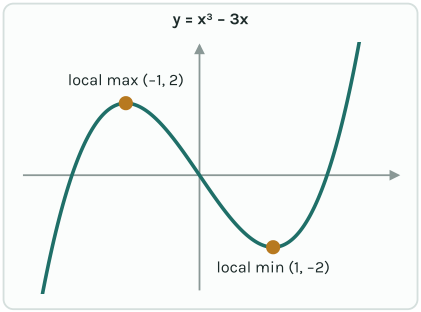
Line	Equation at $x = a$
Tangent	$y - f(a) = f'(a)(x - a)$
Normal (⊥ tangent)	$y - f(a) = -1/f'(a) \cdot (x - a)$
If $f'(a) = 0$	tangent $y = f(a)$, normal $x = a$

EXAMPLE $y = x^2$ at $x = 3$: point (3, 9), gradient $f'(3) = 6$.
Tangent: $y - 9 = 6(x - 3)$ → **$y = 6x - 9$**
Normal: $y - 9 = -\frac{1}{6}(x - 3)$ → **$y = -x/6 + 19/2$**
Check: $6 \times (-\frac{1}{6}) = -1$ ✓



Stationary points: $f'(x) = 0$ — then test the sign

Sign of f' (left, at, right)	Shape	Type
+ 0 −	↗→↘	local maximum
− 0 +	↘→↗	local minimum
+ 0 +	↗→↗	stationary point of inflection
− 0 −	↘→↘	stationary point of inflection



EXAMPLE $y = x^3 - 3x$: $f'(x) = 3x^2 - 3 = 0$ → $x = \pm 1$. $f''(x) = 6x$: $f''(-1) = -6 < 0$ → **max (−1, 2)**; $f''(1) = 6 > 0$ → **min (1, −2)**.
Strictly increasing on $(-\infty, -1) \cup (1, \infty)$, strictly decreasing on $(-1, 1)$.

Second derivative	Means
$f'' > 0$	concave up ∪ ("holds water") → a stationary point here is a min
$f'' < 0$	concave down ∩ ("spills water") → a stationary point here is a max
f'' changes sign	**point of inflection** (e.g. $y = x^3 - 3x^2$ at (1, −2))
$f''(a) = 0$	**no conclusion** — use the sign test

TRAP

$f''(a) = 0$ proves nothing

$y = x^4$ has $f''(0) = 0$ at a minimum; $y = x^3$ has $f''(0) = 0$ at an inflection. Same f'', different answers — always fall back on the sign table.

Absolute max / min on a closed interval

$f(x) = x^3 - 3x$ on $[-2, 3]$ — check stationary points AND endpoints				
x	−2 (end)	−1 (stat.)	1 (stat.)	3 (end)
f(x)	−2	2	−2	**18**

Absolute max **18** at x = 3 (an endpoint!). Absolute min **−2** at x = −2 and x = 1.

Optimisation — the 6-step recipe

1 Write the quantity → 2 Use the constraint to get one variable → 3 State the domain → 4 Solve f′ = 0 → 5 Justify max/min → 6 Answer in context + units

EXAMPLE Open box, square base s, volume 32 cm³. Height $h = 32/s^2$, so surface $S = s^2 + 4sh = s^2 + 128/s$, domain $s > 0$. $S' = 2s - 128/s^2 = 0 \to s^3 = 64 \to s = 4$. $S''(4) = 2 + 256/64 = 6 > 0 \to$ minimum.
Minimum surface area 48 cm² when the base is 4 cm.

Rates of change and related rates

Idea	Rule
Related rates (chain rule)	`dV/dt = dV/dr × dr/dt`
Sphere / circle	`dV/dr = 4πr², dA/dr = 2πr`
Reciprocal	`dt/dx = 1 ÷ (dx/dt)`

EXAMPLE A sphere is inflated at 100 cm³/s. How fast is r growing when r = 5?
dr/dt = (dV/dt) ÷ (dV/dr) = 100 ÷ (4π × 25) = **1/π cm/s**. (Differentiate first, substitute r = 5 last.)

Try it yourself — Section 7

1. Find the tangent to $y = x^3$ at x = 1.
2. Find and classify the stationary points of $y = x^3 - 6x^2 + 9x$.
3. Find the absolute maximum of $f(x) = x^2 - 4x$ on $[0, 5]$.

8 Integration

Standard antiderivatives (add + c)

f(x)	∫ f(x) dx
`xⁿ (n ≠ −1)`	`xⁿ⁺¹/(n + 1)`
`(ax + b)ⁿ`	`(ax + b)ⁿ⁺¹ / (a(n + 1))`
`1/x`	`logₑ|x|`
`1/(ax + b)`	`(1/a) logₑ|ax + b|`
`f′(x)/f(x)`	`logₑ|f(x)|`

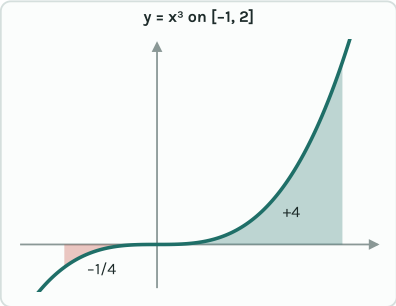
f(x)	∫ f(x) dx
`e^(kx)`	`e^(kx)/k`
`sin(kx)`	`−cos(kx)/k`
`cos(kx)`	`sin(kx)/k`
`1/cos²(kx)`	`tan(kx)/k`
`1/√x`	`2√x`

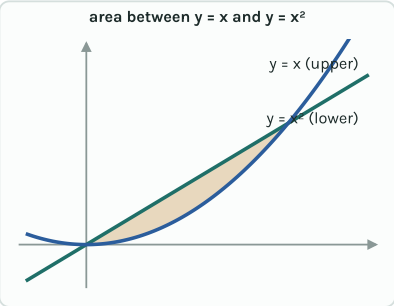
EXAMPLE $\int (3x+1)^7\,dx = (3x+1)^8/24 + c$ · $\int 1/(2x+3)\,dx = \tfrac{1}{2}\log_e|2x+3| + c$ · $\int x/(x^2+1)\,dx = \tfrac{1}{2}\log_e(x^2+1) + c$

CHECK
Differentiate your answer back
You must get the integrand exactly. This catches a missing 1/a or a lost minus sign from sin in seconds.

Definite integrals and area

Fact	Rule
Fundamental theorem	$\int_a^b f(x)\,dx = F(b) - F(a)$ (no + c)
Swap terminals	$\int_a^b f = -\int$ (from b to a) f
Odd function on [-a, a]	$\int = 0$ (e.g. $\int_{-2}^{2} x^3\,dx = 0$)
Area below the x-axis	counts as **negative** in the integral
Area between curves	$\int$ (upper − lower) dx
Average value	$1/(b-a) \times \int_a^b f(x)\,dx$ (x^2 on [0, 3] → 3)





EXAMPLE $y = x^3$ on [−1, 2]: the integral gives −1/4 + 4 = 15/4, but the **area** is 1/4 + 4 = **17/4**.
Between $y = x$ and $y = x^2$: they meet at $x = 0, 1 \rightarrow \int_0^1 (x - x^2)\,dx = 1/2 - 1/3 =$ **1/6**.

TRAP

Integral ≠ area

$\int_0^{2\pi} \sin x\,dx = 0$, but the area is 4. If the curve crosses the axis, find the zeros, integrate each piece, and add the absolute values. An area is never negative.

Kinematics

x position (displacement) ⇄ v velocity = dx/dt ⇄ a acceleration = dv/dt

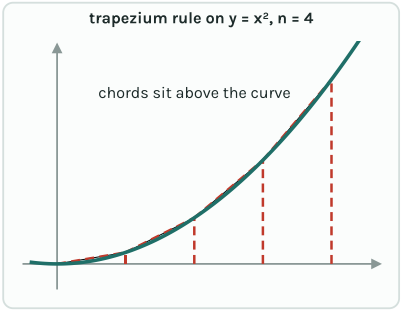
Differentiate going → right, integrate going ← left (and find + c from the starting values). "At rest" means v = 0; "at the origin" means x = 0.

EXAMPLE $v = 3t^2 - 12t$ on [0, 5]. v = 0 at t = 0 and t = 4, so split there.
$\int_0^4 v\,dt = -32$, $\int_4^5 v\,dt = 7$. Displacement = −25, but **distance = 32 + 7 = 39**.

Estimating an integral

$\int_0^2 x^2\,dx = 8/3 \approx 2.667$, with n = 4 (h = 0.5)	
Left rectangles	1.75 (under, x^2 increasing)
Right rectangles	3.75 (over)
Trapezium	$(h/2)[f_0 + 2f_1 + 2f_2 + 2f_3 + f_4] = 2.75$

Trapezium = (left + right)/2. Concave up → trapezium **overestimates**; concave down → underestimates.



TRICK

Unknown terminal or + c

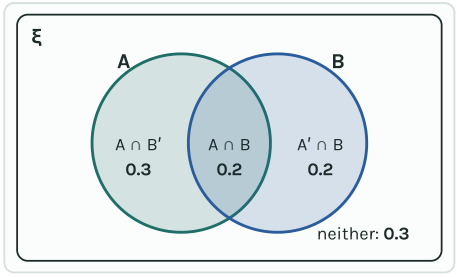
$\int_0^k 2x\,dx = 9 \rightarrow k^2 = 9 \rightarrow k = 3$ ($k > 0$). $f'(x) = 6x$, $f(2) = 5 \rightarrow f(x) = 3x^2 + c$, $12 + c = 5 \rightarrow$ **$f(x) = 3x^2 - 7$**.

Try it yourself — Section 8

1. Find: (a) $\int (4x^3 - 2/x)\,dx$ (b) $\int e^{-2x}\,dx$ (c) $\int \cos(3x)\,dx$
2. Find the area enclosed by $y = x^2 - 4$ and the x-axis.
3. Find the average value of $y = 2x + 1$ on $[0, 4]$.

9 Probability rules and conditional probability

Rule	Formula
Complement	$Pr(A') = 1 - Pr(A)$
Addition ("or")	$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$
Mutually exclusive	$Pr(A \cap B) = 0$
Conditional ("given")	$Pr(A \mid B) = Pr(A \cap B) / Pr(B)$
Multiplication ("and")	$Pr(A \cap B) = Pr(A \mid B) \times Pr(B)$
Independent	$Pr(A \cap B) = Pr(A) \times Pr(B)$
Neither	$Pr(A' \cap B') = 1 - Pr(A \cup B)$



Pr(A) = 0.5, Pr(B) = 0.4, Pr(A ∩ B) = 0.2

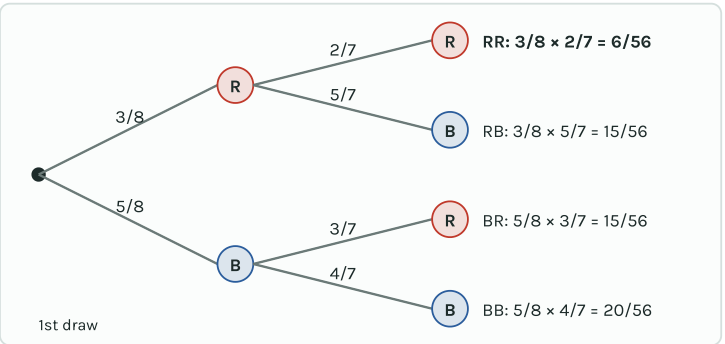
The two-way (Karnaugh) table — fastest method for "and / or / given / neither"

	B	B′	Total
A	**0.2**	0.5 − 0.2 = **0.3**	0.5
A′	0.4 − 0.2 = **0.2**	**0.3**	0.5
Total	0.4	0.6	**1**

EXAMPLE Pr(A ∪ B) = 0.5 + 0.4 − 0.2 = **0.7**. Neither = **0.3**. Pr(A | B) = 0.2 ÷ 0.4 = **0.5** = Pr(A) → A and B are **independent**. (Check: 0.5 × 0.4 = 0.2 ✓)

Fill the totals you are given, then the inner cells. Every row and column must add up, and the corner is always 1.

Tree diagrams — multiply along, add across



3 red and 5 blue counters, draw 2 **without** replacement

	With replacement	Without replacement
Pr(both red)	$3/8 \times 3/8 = 9/64$	$3/8 \times 2/7 = 3/28$
Probabilities	stay the same	change each draw
Model	binomial ✓	tree, **not** binomial

Pr(one of each) = RB + BR = 15/56 + 15/56 = 30/56 = 15/28.
Branches leaving any point must add to 1.

TRAP

Mutually exclusive is NOT the same as independent

Mutually exclusive: A and B **can't both happen** (Pr(A ∩ B) = 0). Independent: one **doesn't change** the chance of the other. If both events can happen on their own, mutually exclusive events are always **dependent**. Always prove independence with numbers: compare Pr(A ∩ B) with Pr(A) × Pr(B).

Try it yourself — Section 9

1. Pr(A) = 0.6, Pr(B) = 0.5 and Pr(A ∩ B) = 0.3. Find Pr(A ∪ B) and Pr(A′ ∩ B′).
2. Are A and B in Q1 independent? Show why.

10 Discrete random variables and the binomial distribution

LINK TO YEAR 7 — COLLECTING & SUMMARISING DATA

E(X) is the "mean from a frequency table" you already know

In Year 7: mean = Σ(score × frequency) ÷ total. Divide each frequency by the total and you get a probability — so the mean becomes $E(X) = \Sigma\, x \cdot Pr(X = x)$.

Books read, x	0	1	2	3	Total
Frequency	6	6	4	1	17
Pr(X = x)	6/17	6/17	4/17	1/17	1

Year 7: (0×6 + 1×6 + 2×4 + 3×1) ÷ 17 = 17 ÷ 17 = 1. Methods: 0(6/17) + 1(6/17) + 2(4/17) + 3(1/17) = **1**. Same answer!

A probability distribution — the one-table method

Two rules: every probability is between 0 and 1, and they **add to 1**. Then build E(X) and E(X²) in the same table:

x	0	1	2	3	Sum
Pr(X = x)	0.1	0.3	**k = 0.4**	0.2	1 → k = 0.4
x · Pr	0	0.3	0.8	0.6	**E(X) = 1.7**
x² · Pr	0	0.3	1.6	1.8	**E(X²) = 3.7**

$Var(X) = E(X^2) - [E(X)]^2 = 3.7 - 1.7^2 = 0.81 \rightarrow sd(X) = \sqrt{0.81} = 0.9$

TRAP

E(X²) is not [E(X)]²

Square each **x**, keep the probabilities the same. If your variance comes out negative, you squared the wrong thing.

Changing X: aX + b

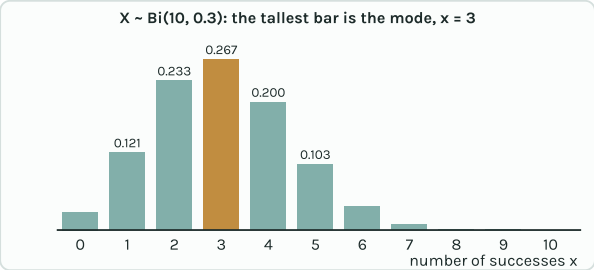
Measure	Rule	E(X) = 2.1, Var(X) = 0.49 → for 3X − 1
Mean	$E(aX + b) = aE(X) + b$	3(2.1) − 1 = **5.3**
Variance	$Var(aX + b) = a^2\, Var(X)$	9 × 0.49 = **4.41** (the −1 does nothing)
Standard deviation	$sd(aX + b) = \lvert a \rvert\, sd(X)$	3 × 0.7 = **2.1**

The binomial distribution X ~ Bi(n, p)

Use it only when **all four** are true:

n	**2**	**p**	**⊥**
fixed number of trials	outcomes: success / failure	same chance every trial	trials independent

What	Formula
Probability	$Pr(X = x) = {}^nC_x\, p^x\, (1 - p)^{n-x}$
Mean	$E(X) = np$
Variance	$Var(X) = np(1 - p)$
At least one	$Pr(X \geq 1) = 1 - (1 - p)^n$
Most likely value	mode = whole-number part of $(n + 1)p$



(n + 1)p = 11 × 0.3 = 3.3 → mode 3. Mean np = 3, variance 2.1

EXAMPLE X ~ Bi(5, 0.2): $Pr(X = 2) = {}^5C_2(0.2)^2(0.8)^3 = 10 \times 0.04 \times 0.512 =$ **0.2048**.
Check: the powers 2 + 3 add to n = 5 ✓

Turning words into inequalities

The question says…	Write	Values included (x = 0 to 6)	Calculator form
exactly 3	`Pr(X = 3)`	0 1 2 **3** 4 5 6	`binomPdf(n, p, 3)`
at most 3 / no more than 3	`Pr(X ≤ 3)`	**0 1 2 3** 4 5 6	`binomCdf(n, p, 0, 3)`
fewer than 3 / less than 3	`Pr(X ≤ 2)`	**0 1 2** 3 4 5 6	`binomCdf(n, p, 0, 2)`
at least 3 / no fewer than 3	`Pr(X ≥ 3) = 1 − Pr(X ≤ 2)`	0 1 2 **3 4 5 6**	`binomCdf(n, p, 3, n)`
more than 3	`Pr(X ≥ 4)`	0 1 2 3 **4 5 6**	`binomCdf(n, p, 4, n)`
between 2 and 5 inclusive	`Pr(2 ≤ X ≤ 5)`	0 1 **2 3 4 5** 6	`binomCdf(n, p, 2, 5)`

TRAP

Off by one in "at least"

For a discrete X: `Pr(X ≥ 3) = 1 − Pr(X ≤ 2)`, **not** 1 − Pr(X ≤ 3). For Bi(20, 0.4) the right answer is 0.9964; the wrong one gives 0.9840 — close enough to look fine, still wrong.

TRICK

"At least one" is one line

Never add Pr(X = 1) + Pr(X = 2) + … Use `1 − (1 − p)ⁿ`. With n = 5, p = 0.2: 1 − 0.8^5 = **0.67232**.
Backwards — "how many trials for at least a 95% chance?": 1 − 0.8^n ≥ 0.95 → n ≥ 13.4 → **n = 14** (always round **up**; n = 13 only gives 0.945).

TRICK

Conditional binomial: divide, don't subtract

`Pr(X ≥ 2 | X ≥ 1) = Pr(X ≥ 2) ÷ Pr(X ≥ 1)` because "X ≥ 2" sits inside "X ≥ 1". For Bi(8, 0.25): 0.6329 ÷ 0.8999 = **0.7033**.

CHECK

Binomial variance is always smaller than the mean

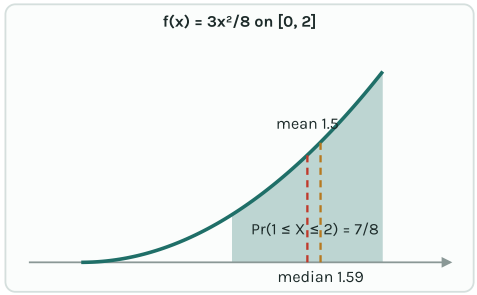
np(1 − p) < np. Bi(20, 0.4): mean 8, variance 4.8. If you get a variance of 8 or more, you forgot the (1 − p).

Try it yourself — Section 10

1. x = 1, 2, 3 with Pr = 0.2, k, 0.5. Find k and E(X).
2. X ~ Bi(10, 0.1). Find E(X), Var(X) and Pr(X ≥ 1).
3. Write "more than 2 but fewer than 6" as an inequality in X.

11 Continuous random variables (probability density functions)

What	Formula
It is a pdf if	`f(x) ≥ 0 and total area = 1`
Probability = area	`Pr(a ≤ X ≤ b) = ∫ₐᵇ f(x) dx`
Mean	`E(X) = ∫ x f(x) dx`
Variance	`E(X²) − [E(X)]², E(X²) = ∫ x² f(x) dx`
Median m	`∫ from start to m of f(x) dx = 0.5`
Mode	x where f(x) is highest (can be an endpoint)



For a continuous X, Pr(X = a) = 0, so ≤ and < give the same answer.

EXAMPLE $f(x) = 3x^2/8$ on $[0, 2]$.
Area: $[x^3/8]$ from 0 to 2 = 8/8 = 1 ✓ · $\Pr(1 \le X \le 2) = 8/8 - 1/8 = \mathbf{7/8}$
$E(X) = \int 3x^3/8\,dx = [3x^4/32] = 48/32 = \mathbf{3/2}$ · $E(X^2) = [3x^5/40] = 96/40 = 12/5$ · $\text{Var} = 12/5 - 9/4 = \mathbf{3/20}$
Median: $m^3/8 = 1/2 \to m^3 = 4 \to \mathbf{m = \sqrt[3]{4} \approx 1.587}$. (Mean ≠ median because the graph is not symmetric.)

TRICK

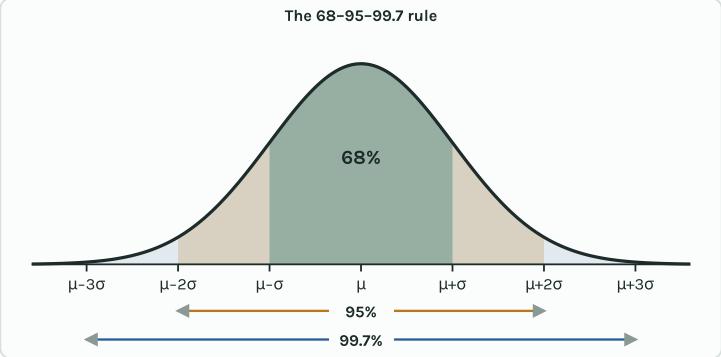
Unknown constant? Set the area to 1

If $f(x) = kx$ on $[0, 4]$, then $\int kx\,dx = 8k = 1$, so $k = 1/8$. Then check k makes $f(x) \ge 0$.

Try it yourself — Section 11

1. $f(x) = x/2$ on $[0, 2]$. Show it is a pdf, then find $\Pr(X < 1)$ and $E(X)$.

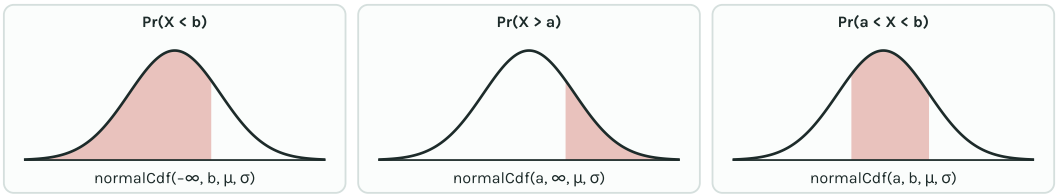
12 The normal distribution $X \sim N(\mu, \sigma^2)$



Fact	What to remember
Shape	bell-shaped, symmetric about μ → mean = median = mode = μ
Notation	`N(50, 9) means σ² = 9, so σ = 3 (not 9!)`
z-score	`z = (x − μ) / σ — how many sds above the mean`
Back to x	`x = μ + zσ`
One tail	$\Pr(X > \mu + \sigma) \approx 0.16$, $\Pr(X > \mu + 2\sigma) \approx 0.025$ (halve what is left over)
Exact 95% cut	$z = \pm 1.96$ (not ±2)

EXAMPLE — COMPARE WITH Z-SCORES Maths: you got 75, class μ = 65, σ = 5 → z = (75 − 65)/5 = **2**.
English: you got 80, class μ = 70, σ = 8 → z = (80 − 70)/8 = **1.25**.
The Maths mark is more impressive — it is further above its mean in standard deviations.

Calculator commands — always sketch and shade first



You want	Command	Example, $X \sim N(70, 4^2)$
a probability	`normalCdf(lower, upper, μ, σ)`	$\Pr(X > 75)$ = normalCdf(75, ∞, 70, 4) = **0.1056**
an x-value from a probability	`invNorm(area to the LEFT, μ, σ)`	$\Pr(X < k) = 0.9 \to k$ = invNorm(0.9, 70, 4) = **75.13**
upper-tail k	`Pr(X > k) = q → invNorm(1 − q, μ, σ)`	$\Pr(X > k) = 0.05$ → enter **0.95**, not 0.05

TRICK

Use symmetry — three answers from one calculation

65 and 75 are both 5 (= 1.25σ) away from 70. So once you know Pr(X > 75) = 0.1056, you also know Pr(X < 65) = **0.1056** and Pr(65 < X < 75) = 1 − 2(0.1056) = **0.7887**.

EXAMPLE — FIND AN UNKNOWN M σ = 5 and Pr(X < 50) = 0.025. Find μ.

① z = invNorm(0.025) = −1.96 ② 50 = μ + (−1.96)(5) ③ μ = 50 + 9.8 = **59.8**.

Sense check: a small left-tail probability means 50 is **below** μ ✓

TRAP

The calculator wants σ, not σ²

X ~ N(50, 9): type σ = **3**. Write "σ = ..." on the page before you touch the calculator.

TRAP

Standard deviations don't add — variances do

For two independent X_1, X_2 with sd σ, $\text{Var}(X_1 + X_2) = 2\sigma^2$, so sd = σ√2. But 2X (one value doubled) has sd 2σ.

More normal-distribution tools

Tool	Rule	Example
Quartiles and IQR	`Q1, Q3 = μ ∓ 0.6745σ; IQR = 1.349σ`	N(100, 15²): Q1 = 89.88, Q3 = 110.12, IQR = 20.23
Normal approximation to a binomial	`Bi(n, p) ≈ N(np, np(1 − p)) if np ≥ 10 and n(1 − p) ≥ 10`	Bi(50, 0.1) fails: np = 5
Transforming X	`aX + b ~ N(aμ + b, a²σ²)`	N(5, 2²) under 3X + 2 → N(17, 6²)

EXAMPLE — M AND Σ BOTH UNKNOWN Pr(X < 40) = 0.1 and Pr(X > 70) = 0.05. Find μ and σ.

z-values: invNorm(0.1) = −1.2816, invNorm(0.95) = 1.6449.

40 = μ − 1.2816σ and 70 = μ + 1.6449σ. Subtract: 30 = 2.9265σ → **σ ≈ 10.25**, then **μ ≈ 53.14**.

Try it yourself — Section 12

1. X ~ N(100, 15²). Use the 68–95–99.7 rule to estimate Pr(X > 130).
2. X ~ N(100, 15²). Pr(X > k) = 0.1. What number do you enter into invNorm?

13 Sample proportions and confidence intervals

LINK TO YEAR 7 — COLLECTING & SUMMARISING DATA

Census vs sample → p vs p̂

	Population (census)	Sample
Who	everyone	a group chosen from the population
Proportion	`p — fixed, usually unknown`	`p̂ = x / n — changes from sample to sample`
Year 7 example	shoe sizes of *everyone* in your team	favourite cereal of 500 Australians

Remember the Year 7 skate-park question: surveying 20 people **at** the skate park is a **biased** sample. A bigger n can't fix a biased method — it must be a **random** sample.

The distribution of $\hat{P}$

What	Formula	Example: p = 0.4, n = 100
Mean	$E(\hat{P}) = p$	0.4
Standard deviation	$sd(\hat{P}) = \sqrt{p(1-p)/n}$	$\sqrt{(0.24/100)}$ = **0.0490**
Approx. normal when	$np \geq 10$ and $n(1-p) \geq 10$	40 ✓ and 60 ✓

TRICK

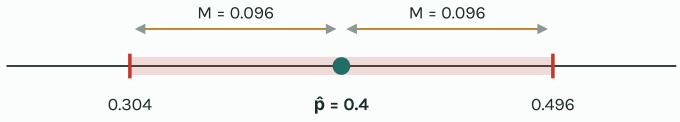
Small n? Turn $\hat{P}$ back into a count

n = 4, p = 0.6: $Pr(\hat{P} \geq 0.5) = Pr(X \geq 2)$ where $X \sim Bi(4, 0.6)$ = **0.8208**. (Multiply $\hat{p}$ by n to get the count.) Don't use the normal here — np = 2.4 is far below 10.

Approximate confidence interval for p

$\left(\hat{p} - z\sqrt{\hat{p}(1-\hat{p})/n} \;,\; \hat{p} + z\sqrt{\hat{p}(1-\hat{p})/n} \right)$ where the margin of error $M = z\sqrt{\hat{p}(1-\hat{p})/n}$

Confidence level	90%	95%	99%
z	1.645	1.96	2.576
Interval	narrower	→	wider



EXAMPLE $\hat{p}$ = 0.4, n = 100, 95%. M = 1.96 × √(0.4 × 0.6 / 100) = 1.96 × 0.0490 = **0.096**.
Interval: 0.4 ± 0.096 = **(0.304, 0.496)**.
Backwards: from an interval, $\hat{p}$ = (upper + lower)/2 and M = (upper − lower)/2.

How big a sample?

Rule	Formula	Example
sample size for margin M	$n \geq z^2 \hat{p}(1-\hat{p}) / M^2$	95%, M ≤ 0.03, no estimate of p: $n \geq 1.96^2 / (4 \times 0.03^2) = 1067.1 \rightarrow$ **n = 1068**
no estimate of p (worst case)	use $\hat{p}(1-\hat{p}) = 0.25$	
halve M	multiply n by **4** (to cut M to a third, multiply n by 9). Doubling n only divides M by √2 ≈ 1.41	

TRAP

Round the sample size UP — always

n ≥ 1067.1 means 1068, even though .1 would normally round down. n = 1067 gives M = 0.030001, which is too big.

TRAP

Say what a confidence interval means — correctly

✓ Correct	"We are 95% confident that the interval (0.304, 0.496) contains p." Or: "about 95% of intervals made this way would contain p."
× Wrong	"There is a 95% chance that p is between 0.304 and 0.496." — p is a fixed number; it is the **interval** that changes between samples.

CHECK

Three checks on any interval

① $\hat{p}$ is exactly the middle. ② Both ends are between 0 and 1. ③ On the same data, 99% is wider than 95%, which is wider than 90%.

Try it yourself — Section 13

1. In a random sample of 400 people, 120 like a new app. Find $\hat{p}$ and an approximate 95% confidence interval for p.
2. A poll wants a margin of error of 0.05 at 95% confidence with no estimate of p. What sample size is needed?
3. If the sample size is multiplied by 4, what happens to the margin of error?

14 Algebra: polynomials and simultaneous equations

Remainder and factor theorems

Theorem	Rule	Example: $P(x) = 2x^3 - 5x^2 + 3x - 7$
Remainder	$P(x) \div (x - a)$ leaves $P(a)$	$\div (x - 2)$: $P(2) = 16 - 20 + 6 - 7 = \mathbf{-5}$
Divisor $ax - b$	remainder $= P(b/a)$	$\div (2x - 1)$: use $P(\frac{1}{2})$, not $P(2)$
Factor	$(x - a)$ is a factor $\Leftrightarrow P(a) = 0$	"$(x + 3)$ is a factor" means $P(-3) = 0$

EXAMPLE Factorise $2x^3 + 3x^2 - 8x + 3$. Coefficients add to 0 → $P(1) = 0$, so $(x - 1)$ is a factor.
Divide: $2x^3 + 3x^2 - 8x + 3 = (x - 1)(2x^2 + 5x - 3) = \mathbf{(x - 1)(2x - 1)(x + 3)}$ → roots 1, ½, −3.

TRICK

Try $x = 1$ and $x = -1$ first

If the coefficients add to 0, then $x = 1$ is a root. If the coefficients with alternating signs add to 0, $x = -1$ is a root. Otherwise try ± factors of (constant ÷ leading coefficient).

Sum and product of roots — answer without solving

Equation	Sum of roots	Product of roots	For $2x^2 - 7x + 3$
$ax^2 + bx + c = 0$	$\alpha + \beta = -b/a$	$\alpha\beta = c/a$	sum 7/2, product 3/2
Useful re-writes: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$, $1/\alpha + 1/\beta = (\alpha + \beta)/(\alpha\beta)$			37/4 and 7/3

Line meets curve — count the meetings with Δ

$\Delta = b^2 - 4ac$	Line and curve…
$\Delta > 0$	meet twice
$\Delta = 0$	touch once — **tangent**
$\Delta < 0$	don't meet

EXAMPLE $y = mx + 2$ and $y = x^2 + x + 3$: $x^2 + (1 - m)x + 1 = 0$, $\Delta = (1 - m)^2 - 4$.
Tangent when $\Delta = 0$ → **$m = -1$ or $m = 3$**.

Simultaneous linear equations with a parameter

For $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, the determinant is $\det = a_1b_2 - a_2b_1$.

det	Lines are…	Solutions
$\det \neq 0$	crossing (different gradients)	exactly one
$\det = 0$	parallel and different	none
	the same line	infinitely many

EXAMPLE $kx + 4y = k + 2$ and $x + ky = 2$: $\det = k^2 - 4 = 0$ at $k = \pm 2$.
$k = 2$: $2x + 4y = 4$ and $x + 2y = 2$ → same line → **infinitely many**.
$k = -2$: $x - 2y = 0$ and $x - 2y = 2$ → parallel → **no solution**. Unique solution for $k \in \mathbb{R} \setminus \{-2, 2\}$.

TRAP

det = 0 does not automatically mean "no solution"

It means parallel **or** identical. Substitute each value back and decide separately — that step is where the marks are.

Try it yourself — Section 14

1. Find the remainder when $x^3 + 2x^2 - 5$ is divided by $(x + 1)$.
2. $(x - 2)$ is a factor of $x^3 + kx - 4$. Find k.

15 Algorithms and pseudocode

Pseudocode	Means
`s ← s + i`	work out the right side with the current values, then store it in s
`if … then … else … end if`	exactly one branch runs
`for i from 1 to 5`	runs 5 times (a to b runs **b − a + 1** times)
`while i < 5 do`	condition checked **before** each pass — may never run
`print s`	inside the loop = one value per pass; after it = one value

Trace a loop in a table

```
s ← 0
for i from 1 to 5 do
  s ← s + i²
end for
print s
```

i	1	2	3	4	5
s	1	5	14	30	**55**

Sum accumulators start at 0; product accumulators start at 1.

Finding roots: bisection vs Newton's method ($f(x) = x^2 - 5$, root $\sqrt{5} \approx 2.2361$)

Bisection on [2, 3]: keep the half with a sign change		
midpoint m	**f(m)**	**new interval**
2.5	1.25	[2, 2.5]
2.25	0.0625	[2, 2.25]
2.125	−0.484	[2.125, 2.25]
2.1875	−0.215	[2.1875, 2.25]

Width after n steps = $(b - a)/2^n$. Slow but **always** works (needs a sign change).

Newton: $x_{n+1} = x_n - f(x_n)/f'(x_n)$, $x_0 = 2$	
x_1	2.25
x_2	2.236111
x_3	2.2360680 ✓

Very fast, but **fails** if $f'(x_n) = 0$, can cycle, or can find a different root if x_0 is badly chosen.

Other algorithms you need

Algorithm	Key fact	Example
Horner's method	nest the polynomial: multiply by x, add the next coefficient	$2x^3 - 6x^2 + 2x - 1$ at $x = 3$: 2 → 0 → 2 → 5
Linear search	check items one by one (any list)	1000 items → up to 1000 checks
Binary search	list must be **sorted**; check the middle, discard half	1000 items → at most 10 checks
Nested loops	n × n passes = n^2	n = 6 → 36 (triangular j ≤ i → 21)

TRAP

Off-by-one errors

for 3 to 9 runs **7** times · while i < n runs one fewer time than while i ≤ n · n strips need **n + 1** ordinates.

Try it yourself — Section 15

1. p ← 1; for i from 1 to 4 do p ← p × i. What is p at the end?
2. Do one Newton step for $f(x) = x^2 - 3$ from $x_0 = 2$.

16 Exam technique

The question says…	You must…
Show that	show every step from the given information to the printed answer; never start from the answer
Hence	use the previous part — another method can score 0
Hence or otherwise	any method, but "hence" is usually quicker
Exact value	surds, π, e, ln, fractions — **no decimals** (π/6, not 0.5236)
Correct to n decimal places	round only at the very end, show exactly n places
State / write down	no working needed
Explain / justify	a sentence naming the reason ("because f′ > 0 on…")
Sketch	label intercepts, turning points, asymptotes (dashed, with equations), endpoints (open/closed)

Marks and time

1 mark	one step or fact
2 marks	method + answer
3+ marks	lay out each step separately
Pace	≈ 1 mark per 1.5 minutes
Multiple choice	≈ 1 min each; never leave one blank

CAS habits

Radian mode	always, every exam
Define f(x) first	then reuse it for d/dx, solve, ∫
Restrict trig solves	`solve(…) | 0 ≤ x ≤ 2π`
Store values	don't retype long decimals
Write what you typed	e.g. "solve(f′(x) = 0, x)" earns the method mark

TRICK

Multiple choice: substitute a number

Put x = 0, 1 or π/2 into the question and each option — usually only one matches. Or eliminate options using one feature you are sure of (domain, asymptote, sign).

TRAP

The most common avoidable losses

① A decimal where an exact value was asked. ② Rounding too early. ③ Answering a slightly different question (x-value vs maximum value). ④ Forgetting endpoints on a closed interval. ⑤ Missing units. ⑥ Leaving a later part blank — follow-through marks exist, so always attempt it.

CHECK

Before you move on

Substitute your answer back · differentiate your antiderivative · is the probability between 0 and 1? · is the area positive? · did you answer the question that was asked?

17 What the official formula sheet gives you — and what it doesn't

✓ ON the sheet (read it off)
Mensuration: cylinder, cone, pyramid, sphere volume, triangle ½bc sin A
Derivative/antiderivative pairs: x^n, $(ax + b)^n$, e^(ax), $\log_e x$, sin, cos, tan (derivative only)
Product, quotient and chain rules
Newton's method and the trapezium rule
Probability: complement, addition, conditional, E(X), Var(X), binomial, pdf mean/variance
Sample proportions: $\hat{P} = X/n$, $E(\hat{P}) = p$, $sd(\hat{P})$, the confidence interval

× NOT on the sheet (memorise!)
Exact trig values, ASTC, trig graph features, general solutions
Index laws, log laws, change of base
Quadratic formula, discriminant, remainder and factor theorems
Transformation rules, inverse functions, composite rule ran g ⊆ dom f
Tangent/normal equations, stationary point tests, average value, area between curves
Independence test, $z = (x - \mu)/\sigma$, 68–95–99.7, z-values 1.645 / 1.96 / 2.576
$\log_e\|x\|$ (the sheet only gives $\log_e x$ for $x > 0$), bisection, left/right rectangles

CHECK

Get this year's sheet

Download the current sheet from vcaa.vic.edu.au, check it against the lists above, and practise with it beside you. Exam 1 and Exam 2 use the same sheet.

Answers — Try it yourself

Section 1

1. $x \le 4$ and $x \neq -1$ → $(-\infty, -1) \cup (-1, 4]$

2. $(2x+5)/(x+1) = 2 + 3/(x+1)$ → **$x = -1$, $y = 2$**

3. $x = -2$: touches (turning point). $x = 1$: flattens and passes through (stationary point of inflection).

Section 2

1. $2x \in [0, 4\pi]$ → 2 cycles × 2 = **4** solutions

2. Base angle $\pi/3$, cos − in Q2, Q3 → **$x = 2\pi/3, 4\pi/3$**

3. Amplitude 2, period $2\pi/3$, range $[3, 7]$

Section 3

1. ran g = R $\not\subset [0, \infty)$, so no. Restrict to **dom g = $[-2, \infty)$**.

2. x: $1 \div 2 - 1 = -\frac{1}{2}$; y: $-3 + 4 = 1$ → **$(-\frac{1}{2}, 1)$**

3. $\sin(2(x - \pi/6))$: dilation of factor ½ from the y-axis, then translation of $\pi/6$ units in the positive direction of the x-axis.

4. Swap: $x = e^y + 3$ → **$f^{-1}(x) = \log_e(x - 3)$**, domain = ran f = **$(3, \infty)$**

Section 4

1. $m = 8/6$ = **4/3**; midpoint **(1, 5)**; length $\sqrt{(36 + 64)}$ = **10**

2. $m = -1/3$: $y + 1 = -\frac{1}{3}(x - 2)$ → **$y = -x/3 - 1/3$**

Section 5

1. e^(2x) = 4 → **$x = \frac{1}{2} \ln 4 = \ln 2$**

2. $A = e^x$: $(A - 3)(A + 2) = 0$, reject $A = -2$ → **$x = \ln 3$**

3. Asymptote **$y = 2$**, range **$(-\infty, 2)$**

Section 6

1. (a) $12(3x - 2)^3$ (b) $\sin x + x \cos x$ (c) $2x\, e^{(x^2)}$ (d) $1/x$

2. $(2x(x+1) - x^2)/(x+1)^2$ = **$(x^2 + 2x)/(x+1)^2$**

Section 7

1. Point (1, 1), gradient 3 → **$y = 3x - 2$**

2. $y' = 3(x - 1)(x - 3)$: **max (1, 4)**, **min (3, 0)**

3. f(0) = 0, f(2) = −4, f(5) = 5 → absolute max **5** at $x = 5$

Section 8

1. (a) $x^4 - 2\ln|x| + c$ (b) $-\frac{1}{2}e^{(-2x)} + c$ (c) $\frac{1}{3}\sin(3x) + c$

2. Zeros at ±2: $\left|\int_{-2}^{2} (x^2 - 4)\, dx\right|$ = **32/3**

3. $\frac{1}{4}\int_0^4 (2x + 1)\, dx = \frac{1}{4} \times 20$ = **5**

Section 9

1. $\Pr(A \cup B) = 0.6 + 0.5 - 0.3$ = **0.8**; neither = 1 − 0.8 = **0.2**

2. $0.6 \times 0.5 = 0.3 = \Pr(A \cap B)$ → **independent**

Section 10

1. $0.2 + k + 0.5 = 1$ → **$k = 0.3$**; $E(X) = 0.2 + 0.6 + 1.5$ = **2.3**

2. $E(X) = 1$, $\mathrm{Var}(X) = 0.9$, $\Pr(X \ge 1) = 1 - 0.9^{10} \approx$ **0.6513**

3. **$\Pr(3 \le X \le 5)$**

Section 11

1. $f \ge 0$ and $\int_0^2 x/2\, dx = [x^2/4] = 1$ ✓; $\Pr(X < 1)$ = **1/4**; $E(X) = \int_0^2 x^2/2\, dx$ = **4/3**

Section 12

1. $130 = \mu + 2\sigma$ → about 0.025 (exact 0.0228)

2. Left area 1 − 0.1 = **0.9**: invNorm(0.9, 100, 15) ≈ 119.2

Section 13

1. $\hat{p} = 0.3$; $M = 1.96\sqrt{(0.3 \times 0.7/400)} \approx 0.0449$ → **(0.255, 0.345)**

2. $n \ge 1.96^2/(4 \times 0.05^2) = 384.2$ → **$n = 385$**

3. It is **halved** ($M \propto 1/\sqrt{n}$)

Section 14

1. $P(-1) = -1 + 2 - 5$ = **−4**

2. $P(2) = 8 + 2k - 4 = 0$ → **$k = -2$**

Section 15

1. $p = 1 \times 2 \times 3 \times 4$ = **24**

2. $x_1 = 2 - (4 - 3)/(2 \times 2)$ = **1.75**